NAME

mbpy_histplot − Create an executable shellscript which will generate a GMT modern-mode frequency histogram plot of a set of data values.

VERSION

Version 5.0

SYNOPSIS

mbpy_histplot **−I***file* [**−O***root*] [**−−format***fmt*] [**−−title***text*] [**−−xlabel***text*] [**−−ylabel***text*] [**−−dpi***dpi*] [**−X**] [**−V**]
[**−C**[*xmin/xmax/*]*width*] [**−G***color*] [**−W***pen*]
[**−R***xmin/xmax/ymin/ymax*] [**−J***spec*] [**−P***pagesize*] [**−−portrait** | **−−landscape**] [**−−tick−info***spec*]
[**−−gmt−default***param/value*]

DESCRIPTION

mbpy_histplot is a Python/PyGMT-era replacement for the Perl macro **mbm_histplot**. It reads a file containing one data value per line and generates an executable shellscript of **GMT modern mode** commands, built around **gmt histogram** (the modern-mode name for classic **pshistogram**), which produces a frequency histogram of those values when run: the bars show what percentage of the data falls in each value bin, and mean / ±1 standard deviation / ±2 standard deviation normal-distribution curves are always overlaid (in black, red, and green respectively) for reference.

mbpy_histplot shares its page-layout and projection-fitting conventions with **mbpy_grdplot** and **mbpy_grd3dplot** (and, in fact, the underlying **mbpy_layout.py** module), except that its default projection is a plain linear/Cartesian one (matching **mbm_histplot**'s own default of **−Jx**, rather than the Mercator/auto-fit default the grid-plotting macros use) -- appropriate since a histogram's axes are ordinary numeric value/percentage scales, not geographic coordinates.

Unlike **mbm_histplot**'s **−L***title***:***xlabel***:***ylabel* colon-joined label option, **mbpy_histplot** exposes **−−title**/**−−xlabel**/**−−ylabel** as separate options.

mbpy_histplot does not expose a data-rescaling, xy-overlay, or custom CPT-file option: **mbm_histplot** itself refers to variables for all of these ($data_scale, @xyfiles, $file_cpt) that are never actually assigned from any command-line option in its own source -- they are leftover, unreachable copy-paste from the **mbm_grdplot** template **mbm_histplot** was originally cloned from, not real functionality.

Note: classic **pshistogram**'s outline-pen option was **−L***pen*; the current **gmt histogram** module (checked directly against its own built-in help) has no **−L** option at all and uses **−W***pen* instead, like nearly every other GMT plotting module's pen option. **mbpy_histplot**'s own **−W**/ **−−pen** option is written out to the generated script using the current **−W** syntax, not the classic **−L** mbm_histplot's own Perl source still emits (which is simply not accepted by a current GMT installation).

mbpy_histplot requires **GMT** 6 or later (modern mode) and a Python 3 interpreter. It depends on **mbpy_layout.py** (shared with **mbpy_grdplot**/**mbpy_grd3dplot**) and its own **mbpy_histplot_script.py**, installed alongside it.

MB-SYSTEM AUTHORSHIP

David W. Caress
Monterey Bay Aquarium Research Institute
Dale N. Chayes
Center for Coastal and Ocean Mapping

University of New Hampshire
Christian do Santos Ferreira
MARUM - Center for Marine Environmental Sciences
University of Bremen

OPTIONS

−I, **−−input** *file*
Input data file, one numeric value per line. Required.

−O, **−−output** *root*
Output root name. Default: the input file's own basename.

−−format *fmt*
Output format passed to **gmt figure**. Default **pdf**.

−−title *text*
Plot title. Default: "Frequency Histogram of *file*".

−−xlabel *text*
X-axis (value) label. Default: none.

−−ylabel *text*
Y-axis (frequency) label. Default: "Frequency %".

−−dpi *dpi*
The final raster's resolution (via **gmt figure**).

−X, **−−execute**
Run the generated script immediately after writing it.

−V, **−−verbose**
Print the generated script's path and a one-line summary.

−C, **−−cell−width** *[xmin/xmax/]width*
Histogram bin width. If given as three slash-separated values, also overrides the data's own detected x (value) bounds. Default width: (xmax−xmin)/10.

−G, **−−fill** *color*
Bar fill color, or **N** for unfilled (outline-only) bars. Default **gray**.

−W, **−−pen** *pen*
Bar outline pen. Default **1p**.

−R, **−−region** *xmin/xmax/ymin/ymax*
Plot region. Default: the data's own value range for x; a y (frequency-percent) range computed from a pre-scan of the data, with a 10% margin above the tallest bin.

−J, **−−projection** *spec*
GMT projection/scale spec, e.g. **x0.02** (linear, 0.02 inch/unit). Default: an auto-fit linear (Cartesian) projection chosen to fill the page.

−P, **−−pagesize** *size*
Page size code. Default **a**.

−−portrait, **−−landscape**
Force the page orientation instead of choosing automatically.

−−tick−info *spec*
Raw **−B** basemap tick-interval spec, overriding the automatically computed one.

−−gmt−default *param/value*
Override a GMT default parameter for this plot; repeatable.

EXAMPLES

A histogram of a column of survey depth values, with custom bin width, fill color, and labels, rendered directly to a PNG:

```
mbpy_histplot −I depths.txt −O depth_hist \
  −−cell−width 2.5 −−fill orange −−pen "1p,black" \
  −−title "Depth Distribution" −−xlabel "Depth (m)" \
  −−format png −−execute −V
```

An outline-only (unfilled) histogram with an explicit bin range:

```
mbpy_histplot −I values.txt −O values_hist \
  −−cell−width "0/100/5" −−fill N −−pen "2p,blue" \
  −−format pdf −V
```

SEE ALSO

mbm_histplot(1), **mbpy_grdplot**(1), **mbpy_grd3dplot**(1), **mbpy_xyplot**(1), **mbhistogram**(1), **mbdefaults**(1), **mbsystem**(1)

BUGS

Please report any bugs to the MB-System project on GitHub (https://github.com/dwcaress/MB-System).